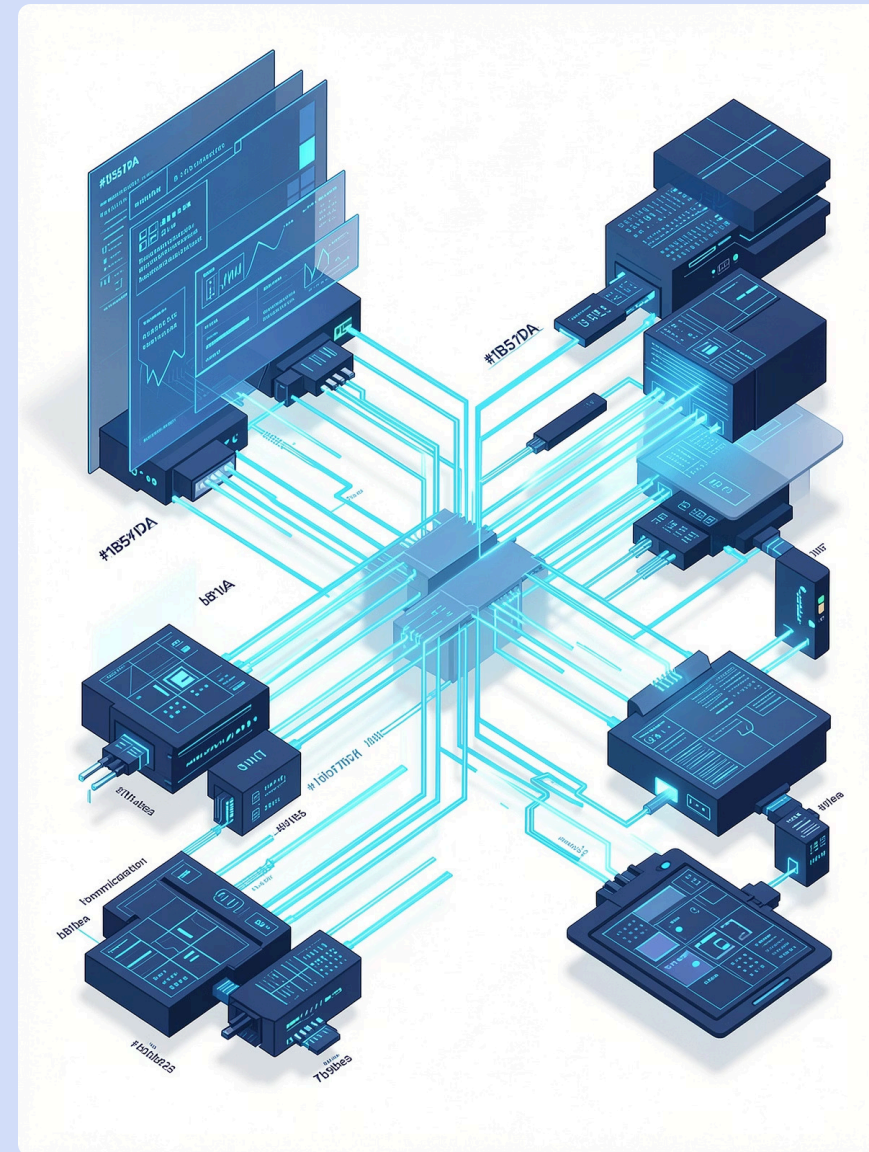


Churn Prediction Service

A production-ready ML service that predicts customer churn for telecom companies – enabling proactive retention before customers leave. Built with FastAPI, CatBoost, Streamlit, and Docker.

V2.0.0 · 2026

IVAN SAZONTOV



Why Churn Prediction Matters

25%

Annual Churn Rate

Average in telecom industry

\$500

Acquisition Cost

Cost to acquire a new customer

\$75

Retention Cost

Cost to retain an existing customer

6.7x

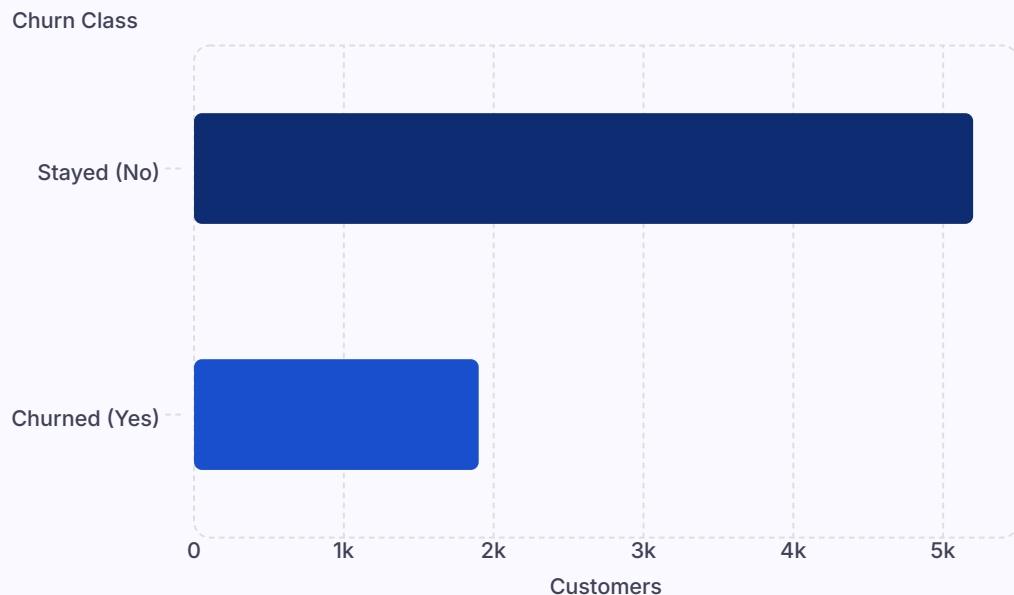
Savings Ratio

Every \$1 on retention saves \$6.70 on acquisition

US telecom loses ~**\$30 billion annually** to churn. The model finds **96 out of 100** customers who will leave – enabling intervention before they do.



Exploratory Data Analysis



Dataset: Telco Customer Churn

- **7,043 records** across 21 features
- **Class imbalance:** 73.5% stayed vs 26.5% churned
- Top churn predictors: month-to-month contracts, fiber optic, electronic check payments, no security add-ons
- Strong correlation (0.83) between `tenure` and `TotalCharges` – one dropped to reduce multicollinearity

i Priority metrics: **Recall**, – accuracy alone would be misleading on imbalanced data.

Preprocessing & Feature Engineering

1

TotalCharges Cleaner

Converts to numeric; applies domain logic: `tenure=0` → `TotalCharges=0`. Zero missing values remain.

2

6 Engineered Features

`avg_monthly_spend`, `tenure_group` (new/mid/loyal),
`has_internet`, `has_addons`, `risky_contract`, `auto_payment`

3

Encoding & Scaling

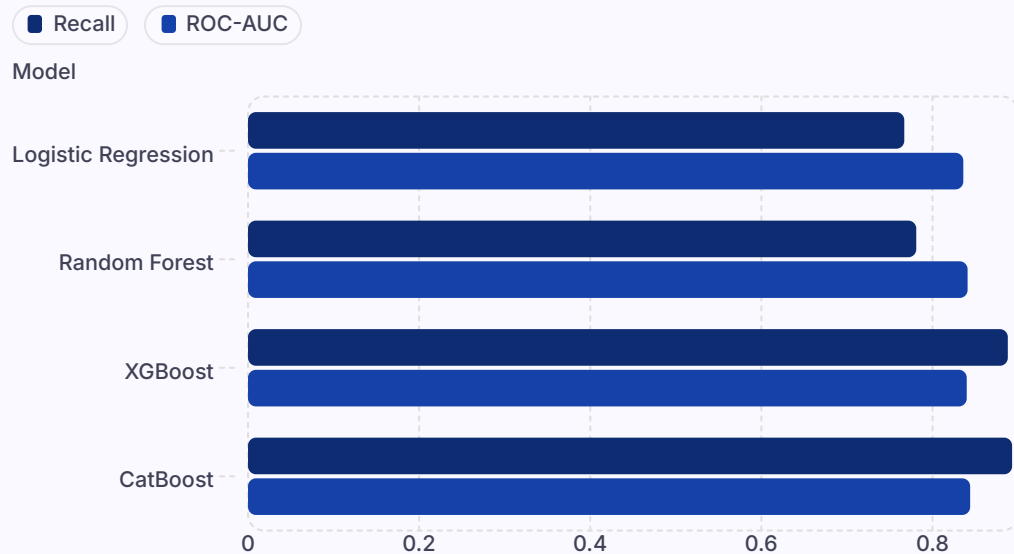
Binary → 0/1; ordinal → integers; nominal → one-hot.
StandardScaler on numerical features only.

4

Feature Selection

Removed 4 redundant features (correlation >0.8). Final set: 23 features, no data leakage.

Model Selection & Performance



CatBoost Selected as Final Model

9 models compared. CatBoost won on **highest recall (89.3%)** and stable generalization.

Recall: 95.72%

Detects ~96 of 100 actual churners

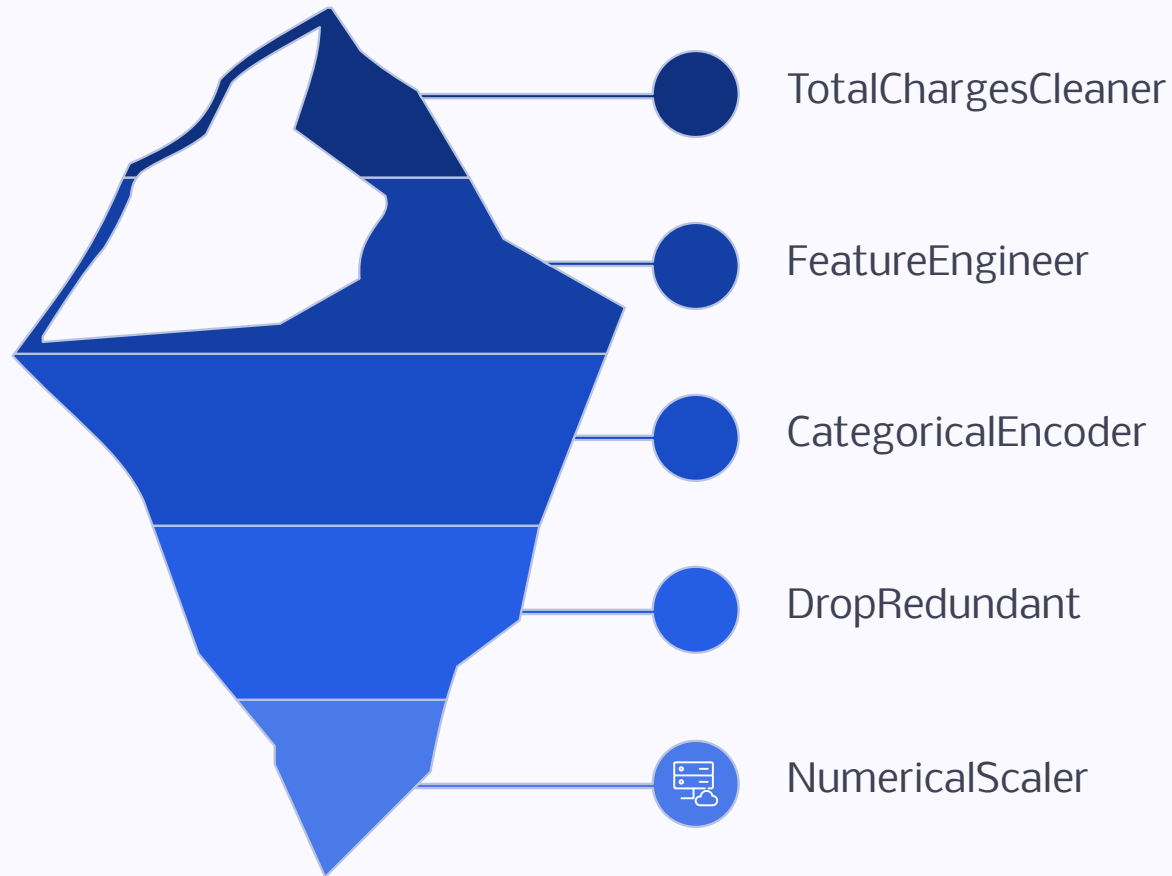
ROC-AUC: 84.35%

Strong discriminative power

F1-Score: 55.29%

Trade-off for high recall – acceptable

ML Pipeline Architecture



Raw customer data flows through 6 automated steps to produce churn predictions in a single call. The pipeline is saved as .pkl (~160 KB) and loaded once at startup for production inference.

REST API & Authentication

🔑 API Key Auth

SHA-256 hashed keys via `X-API-Key` header. Three roles: **user**, **admin**, **dashboard**.

📊 Prediction Endpoints

`POST /predict` (single) and `POST /predict/batch` – returns binary prediction + probability score.

🏷️ Label Management

CRUD + batch endpoints for ground truth collection. Labels stored in SQLite with timestamps.

⚡ Rate Limiting

100 - 1,000 req/min per key. Custom middleware with sliding 60-second window.



Dashboard & Economic Impact



Live Dashboard Metrics

- **223 customers** monitored, 202 labeled (90.6%)
- **MRR at risk:** \$703/month · \$598/month high-risk
- **high-risk customers** flagged for immediate outreach

Model Economic Impact

- **Lost MRR:** \$5,252/month from actual churners
- **Currently saving:** \$1,554/month via model intervention
- **Annual impact:** \$18,648 saved + \$2,316 potential

✓ Top 5 high-risk customers represent **\$488/month → \$5,856/year** at stake. Call them today.

Deployment, Testing & ROI

Docker Architecture

Two containers orchestrated via Docker Compose on Ubuntu VPS:

- **churn-api** – FastAPI on port 8000, 23 endpoints
- **churn-dashboard** – Streamlit on port 8501
- **Shared volumes** persist SQLite DB, models, logs across restarts

Testing

15/15 tests passed (pytest) covering auth, integration, rate limiting, and transformers in 2.72s.

Business ROI

\$4.97M/year

Revenue saved (100K customers, 35% save rate)

33,000% ROI

First-year return on \$15,060 total cost

<1 hour

Payback period from first operation

Summary & Future Roadmap

A complete, production-ready churn prediction service – operational, documented, tested, and deployed live.

- **PostgreSQL Upgrade**
Replace SQLite for better concurrency and horizontal scaling.
- **Monitoring & Alerting**
Prometheus + Grafana for API latency, prediction volume, and model drift with Slack alerts.
- **A/B Testing Framework**
Deploy multiple model versions with traffic splitting to measure real-world retention impact.
- **Customer Segmentation**
clustering + churn probability for personalized retention offers.

❏ **Live Demo:** API <http://207.231.105.113:8000/docs> ·
Dashboard <http://207.231.105.113:8501/> · GitHub:
github.com/greencat1/Portfolio